

Bank Statement Verification System

Overview

Comprehensive verification system for bank statement analysis with 3 main check categories.

Check Categories

1. PDF Integrity Check

Purpose: Validates PDF document authenticity and structural integrity

How::

- Structure integrity check (validates %%EOF markers, startxref, proper start/end)
- Metadata integrity check (compares CreationDate vs ModDate, checks for suspicious producer)
- File size validation (checks if file is empty or too small <100 bytes)
- PDF signature verification (validates %PDF- header and version)
- Cross-reference table validation (validates page access and xref integrity)

2. OFAC Sanctions Check

Purpose: Screens transactions against OFAC sanctions list for compliance

How::

- Fuzzy matching with confidence scoring (SequenceMatcher similarity)
- Embedding-based semantic matching (ultra-fast cosine similarity with sentence-transformers)

3. Fraud Detection

Purpose: Identifies suspicious transaction patterns and potential fraud

How::

- Rules-based detection (13 fraud rules):
 - Credit/debit reconciliation (checks if credits + debits = final balance)
 - Suspicious merchants (gambling, crypto)
 - Large amounts $>3\sigma$ from mean
 - Micro-transactions $<\$10$ (structuring)
 - High-frequency payments to same recipient
 - Large withdrawals $>\$25000$
 - Round-trip transactions (deposit + withdrawal same day)
 - Same-day multiple large transactions
- ML anomaly detection using Isolation Forest (flags unusual transaction amounts with 10% contamination)
- Statistical profiling analysis (calculates mean, median, std dev for amount thresholds)
- Hybrid scoring system (normalizes fraud scores 0-1, flags if score ≥ 0.3)